Dataset Construction

Source data

MASSIVE
dataset
(60 intents)

- English
  - 3k train
  - 2k test
- Bengali
  - 3k train
  - 2k test

Synthetic data

Claude Sonnet 5
generates samples

Human annotators
write and verify

- Code-mixed
  - 3k train
  - 2k test
- Romanized
  - 3k train
  - 2k test

Fine-tune with Three Strategies

(a) Direct fine-tuning

M_0 **Pre-trained model**

English data → **Fine-tune** Qwen3.5-0.8B → M_{En} **English** fine-tuned model

Bengali data → **Fine-tune** Qwen3.5-0.8B → M_{Bn} **Bengali** fine-tuned model

Code-mixed data → **Fine-tune** Qwen3.5-0.8B → M_{CM} **Code-mixed** fine-tuned model

Romanized data → **Fine-tune** Qwen3.5-0.8B → M_{Rom} **Romanized** fine-tuned model

(b) Sequential fine-tuning

M_0 **Pre-trained model**

Fine-tune Qwen3.5-0.8B ← English data → M_1 Intermediary **English model**

Fine-tune Qwen3.5-0.8B ← Bengali data → M_2 Intermediary **Bengali model**

Fine-tune Qwen3.5-0.8B ← Code-mixed data → M_3 Intermediary **Code-mixed model**

Fine-tune Qwen3.5-0.8B ← Romanized data → M_{seq} **Final model** sequential result

(c) Joint fine-tuning

English data
Bengali data
Code-mixed data
Romanized data

→ **Fine-tune** Qwen3.5-0.8B → M_{joint} **Joint model**

Evaluation Protocol

Test sets
En Bn CM Rom

→ **Inference**
score each query against all 60 intents (full label list)

→ **Compare**
zero-shot M_0 vs. fine-tuned

→ **Metrics**